

MATH-810 Mathematical Methods for AI

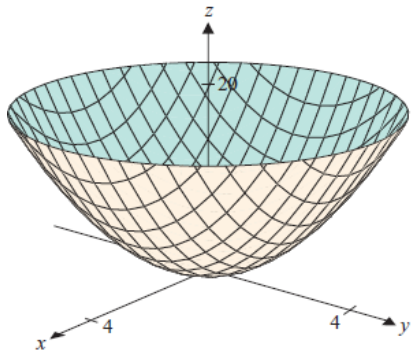
Partial Derivatives, Gradient, Hessian

Dr. Yasir Ali (yali@ceme.nust.edu.pk)

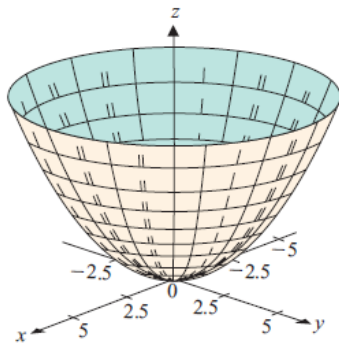
DBS&H, CEME-NUST

October 20, 2025

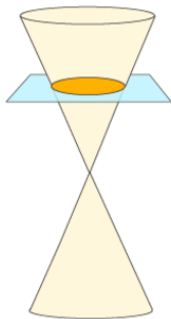




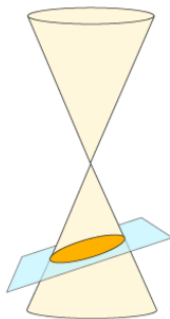
$$z = x^2 + y^2$$



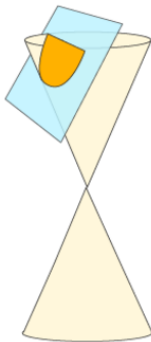
$$z = x^2 + y^2$$



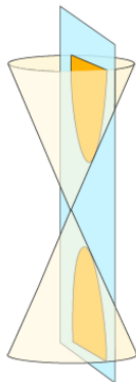
Circle



Ellipse

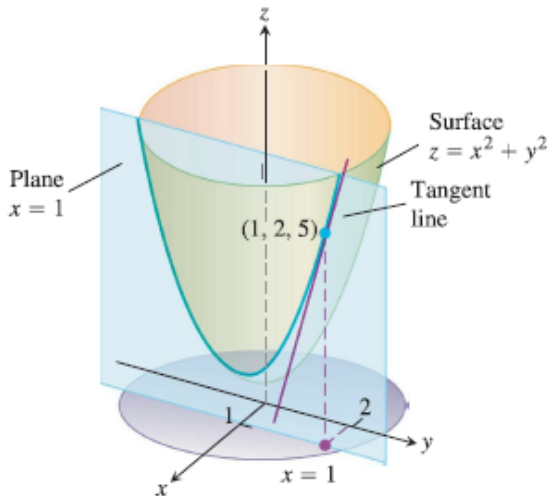


Parabola

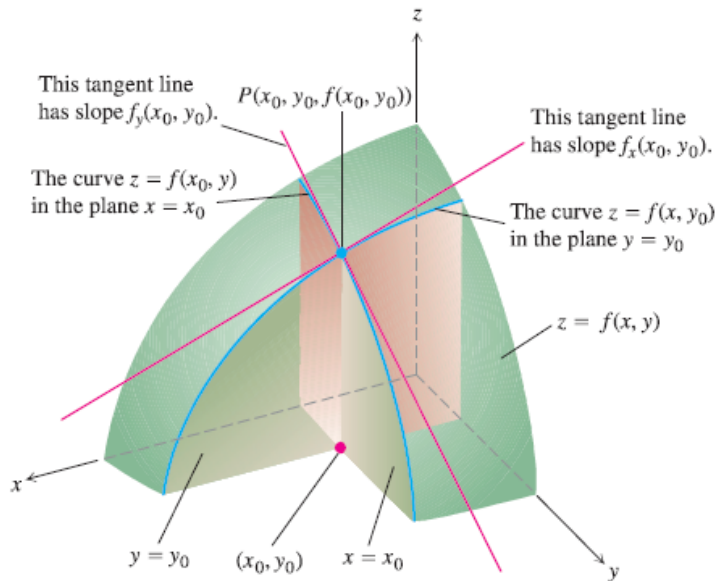


Hyperbola





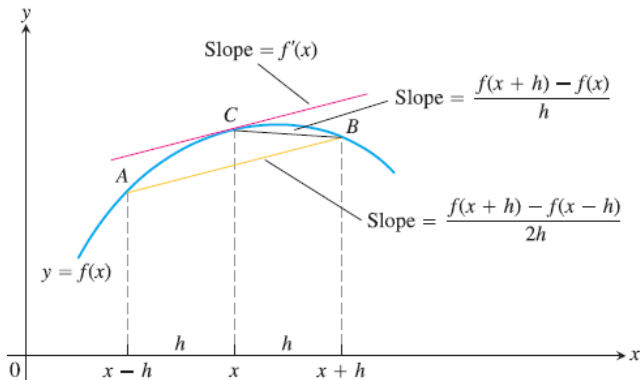
The plane $x = 1$ intersects the paraboloid $z = x^2 + y^2$



Derivatives

Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$



Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along x-axis**

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

Derivative of $f(x)$ at point a **along x-axis**

$$f'(x)|_{\text{at } x=a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along x-axis**

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

Partial Derivative of $f(x, y)$ at point (a, b) **along y-axis**

$$\frac{\partial f}{\partial y}(a, b) = \lim_{k \rightarrow 0} \frac{f(a, b+k) - f(a, b)}{k}$$

Partial Derivatives of $f(x, y)$

Problem: Find $f_x(x, y)$ and $f_y(x, y)$ for

$$f(x, y) = 2x^3y^2 + 2y + 4x$$

and evaluate them at $(x, y) = (1, 3)$.

Solution:

$$f_x(x, y) = \frac{\partial}{\partial x} (2x^3y^2 + 2y + 4x) = 6x^2y^2 + 4, \quad \text{treat } y \text{ as constant}$$

$$f_y(x, y) = \frac{\partial}{\partial y} (2x^3y^2 + 2y + 4x) = 4x^3y + 2. \quad \text{treat } x \text{ as constant}$$

Evaluations:

$$f_x(1, 3) = 6(1)^2(3)^2 + 4 = 58, \quad f_y(1, 3) = 4(1)^3(3) + 2 = 14.$$

Final Results:

$$f_x(1, 3) = 58, \quad f_y(1, 3) = 14$$

Problem: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $z = x^4 \sin(xy^3)$.

Problem: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $z = x^4 \sin(xy^3)$.

Solution:

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial}{\partial x} [x^4 \sin(xy^3)] \\&= x^4 \frac{\partial}{\partial x} [\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial x} [x^4] \\&= x^4 \cos(xy^3) \cdot y^3 + \sin(xy^3) \cdot 4x^3 \\&= x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3).\end{aligned}$$

Problem: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $z = x^4 \sin(xy^3)$.

Solution:

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial}{\partial x} [x^4 \sin(xy^3)] \\&= x^4 \frac{\partial}{\partial x} [\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial x} [x^4] \\&= x^4 \cos(xy^3) \cdot y^3 + \sin(xy^3) \cdot 4x^3 \\&= x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3).\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial y} &= \frac{\partial}{\partial y} [x^4 \sin(xy^3)] \\&= x^4 \frac{\partial}{\partial y} [\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial y} [x^4] \\&= x^4 \cos(xy^3) \cdot 3xy^2 + \sin(xy^3) \cdot 0 \\&= 3x^5 y^2 \cos(xy^3).\end{aligned}$$

Final Results:

$$\frac{\partial z}{\partial x} = x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3), \quad \frac{\partial z}{\partial y} = 3x^5 y^2 \cos(xy^3).$$

Gradient of scalar Function $f(x, y)$

The gradient of $f(x, y)$ is the vector-valued function

$$\nabla f = [f_x, f_y] = f_x \hat{\mathbf{i}} + f_y \hat{\mathbf{j}}$$

The gradient of $f(x, y)$ is the vector-valued function

$$\nabla f = [f_x, f_y, f_z] = f_x \hat{\mathbf{i}} + f_y \hat{\mathbf{j}} + f_z \hat{\mathbf{k}}$$

		TEMPERATURE $T(^{\circ}\text{F})$			
WIND SPEED v (mi/h)		20	25	30	35
	5	13	19	25	31
	10	9	15	21	27
	15	6	13	19	25
	20	4	11	17	24

Use the values of the wind chill index function $W(T, v)$ displayed in Table to estimate the partial derivative of W with respect to v at $(T, v) = (25, 10)$.

$$\frac{\partial W}{\partial v}(25, 10) = \lim_{\Delta v \rightarrow 0} \frac{W(25, 10 + \Delta v) - W(25, 10)}{\Delta v} \approx \frac{W(25, 10 + \Delta v) - 15}{\Delta v}.$$

Case 1: For $\Delta v = 5$,

$$\frac{\partial W}{\partial v}(25, 10) \approx \frac{W(25, 15) - 15}{5} = \frac{13 - 15}{5} = \frac{-2}{5} = -0.4^{\circ}\text{F}/(\text{mi/h}).$$

Case 2: For $\Delta v = -5$,

$$\frac{\partial W}{\partial v}(25, 10) \approx \frac{W(25, 5) - 15}{-5} = \frac{19 - 15}{-5} = \frac{-4}{5} = -0.8^{\circ}\text{F}/(\text{mi/h}).$$

$$\text{Average Approximation: } \frac{\partial W}{\partial v}(25, 10) \approx \frac{-0.4 + (-0.8)}{2} = -0.6^{\circ}\text{F}/(\text{mi/h}).$$

Second-Order Partial Derivatives of $f(x, y) = x^2y^3 + x^4y$

Second-Order Partial Derivatives of $f(x, y) = x^2y^3 + x^4y$

First-Order Partial Derivatives:

$$f_x(x, y) = \frac{\partial}{\partial x} (x^2y^3 + x^4y) = 2xy^3 + 4x^3y,$$

$$f_y(x, y) = \frac{\partial}{\partial y} (x^2y^3 + x^4y) = 3x^2y^2 + x^4.$$

Second-Order Partial Derivatives of $f(x, y) = x^2y^3 + x^4y$

First-Order Partial Derivatives:

$$f_x(x, y) = \frac{\partial}{\partial x} (x^2y^3 + x^4y) = 2xy^3 + 4x^3y,$$

$$f_y(x, y) = \frac{\partial}{\partial y} (x^2y^3 + x^4y) = 3x^2y^2 + x^4.$$

Second-Order Partial Derivatives:

$$f_{xx}(x, y) = \frac{\partial}{\partial x} (f_x(x, y)) = \frac{\partial}{\partial x} (2xy^3 + 4x^3y) = 2y^3 + 12x^2y,$$

$$f_{yy}(x, y) = \frac{\partial}{\partial y} (f_y(x, y)) = \frac{\partial}{\partial y} (3x^2y^2 + x^4) = 6x^2y,$$

$$f_{xy}(x, y) = \frac{\partial}{\partial y} (f_x(x, y)) = \frac{\partial}{\partial y} (2xy^3 + 4x^3y) = 6xy^2 + 4x^3,$$

$$f_{yx}(x, y) = \frac{\partial}{\partial x} (f_y(x, y)) = \frac{\partial}{\partial x} (3x^2y^2 + x^4) = 6xy^2 + 4x^3.$$

Symmetry of Mixed Derivatives:

$$f_{xy}(x, y) = f_{yx}(x, y) = 6xy^2 + 4x^3.$$

Problem: Find the slope of the sphere $x^2 + y^2 + z^2 = 1$ in the y -direction at the points:

$$\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right) \quad \text{and} \quad \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right).$$

Solution:

1. Implicit Differentiation of the Sphere:

$$\frac{\partial}{\partial y} (x^2 + y^2 + z^2) = \frac{\partial}{\partial y} (1).$$

This gives:

$$2y + 2z \frac{\partial z}{\partial y} = 0.$$

2. Solve for $\frac{\partial z}{\partial y}$:

$$\frac{\partial z}{\partial y} = -\frac{y}{z}.$$

3. Evaluate at the Points: - At $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$: $\frac{\partial z}{\partial y} = -\frac{\frac{1}{3}}{\frac{2}{3}} = -\frac{1}{2}.$

- At $\left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right)$: $\frac{\partial z}{\partial y} = -\frac{\frac{1}{3}}{-\frac{2}{3}} = \frac{1}{2}.$

The One-Dimensional Wave Equation

Equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2},$$

where:

- $u(x, t)$ is the unknown function of position x and time t ,
- $c > 0$ is a constant dependent on the physical properties of the string.

Explanation:

- This equation models the motion of a vibrating string.
- It relates the second partial derivatives of u with respect to time and position.
- Since the equation involves partial derivatives of $u(x, t)$, it is classified as a *partial differential equation (PDE)*.

Show that the function $u(x, t) = \sin(x - ct)$ is a solution of Wave Equation

The gradient of $f(x, y)$ is the vector-valued function

$$\nabla f = [f_x, f_y, f_z] = f_x \hat{\mathbf{i}} + f_y \hat{\mathbf{j}} + f_z \hat{\mathbf{k}}$$

Directional Derivative and Gradient of Function

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = f_x u_1 + f_y u_2 + f_z u_3$$

For $f(x, y) = x^2y - 4y^3$, compute $D_{\mathbf{u}}f(2, 1)$ for the directions

① $\mathbf{u} = [\frac{\sqrt{3}}{2}, \frac{1}{2}]$

② $\mathbf{u}$ in the direction from $(2, 1)$ to $(4, 0)$.

$$D_{\mathbf{u}}f = f_x u_1 + f_y u_2, \text{ where } \mathbf{u} = [\overset{u_1}{\frac{\sqrt{3}}{2}}, \overset{u_2}{\frac{1}{2}}]$$

① $f_x = 2xy$ and $f_y = x^2 - 12y^2$. $f_x(2, 1) = 4$, $f_y(2, 1) = -8$

$$D_{\mathbf{u}}f(2, 1) = 4(\frac{\sqrt{3}}{2}) - 8(\frac{1}{2}) = 2\sqrt{3} - 4$$

② unit vector from $(2, 1)$ to $(4, 0)$ is $\mathbf{v} = [\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}]$.

$$D_{\mathbf{v}}f(2, 1) = 4(\frac{2}{\sqrt{5}}) - 8(\frac{-1}{\sqrt{5}}) = \frac{16}{\sqrt{5}}$$

For $f(x, y, z) = x^2y - yz^3 + z$, at the point $(1, -2, 0)$ in the direction of $\mathbf{v} = 2\hat{i} + \hat{j} - 2\hat{k}$

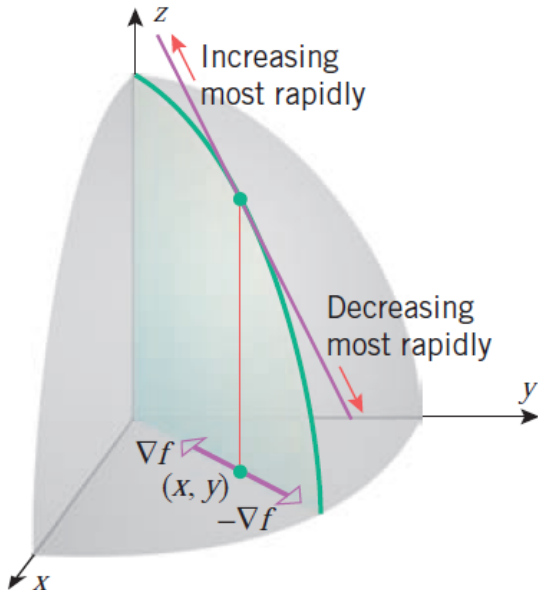
For $f(x, y, z) = x^2y - yz^3 + z$, at the point $(1, -2, 0)$ in the direction of $\mathbf{v} = 2\hat{i} + \hat{j} - 2\hat{k}$

$$D_{\mathbf{u}}f = f_x u_1 + f_y u_2 + f_z u_3, \text{ where } \mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \left[\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right]$$

$$f_x = 2xy, \quad f_y = x^2 - z^3 \text{ and } f_z = -3yz^2 + 1.$$

$$f_x(1, -2, 0) = -4, \quad f_y(1, -2, 0) = 1, \quad f_z(1, -2, 0) = 1$$

$$D_{\mathbf{u}}f(1, -2, 0) = -4\left(\frac{2}{3}\right) + \frac{1}{3} - \frac{2}{3} = -3$$



- 1 The function f increases most rapidly when $\cos \theta = 1$ or when $\theta = 0$ and $\mathbf{u}$ is the direction of ∇f . That is, at each point P in its domain, f increases most rapidly in the direction of the gradient vector ∇f at P . The derivative in this direction is

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos 0 = |\nabla f|$$

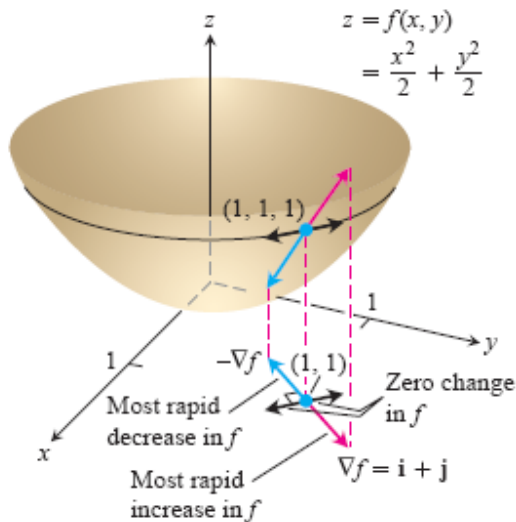
- 2 Similarly, f decreases most rapidly in the direction of $-\nabla f$. The derivative in this direction is

$$D_{\mathbf{u}}f = |\nabla f| \cos \pi = -|\nabla f|$$

- 3 Any direction $\mathbf{u}$ orthogonal to a gradient $\nabla f \neq 0$ is a direction of zero change in f because $\theta = \frac{\pi}{2}$ and

$$D_{\mathbf{u}}f = |\nabla f| \cos \frac{\pi}{2} = 0$$

Example of DD



Directional Derivatives

Let $f(x, y) = x^2 e^y$. Find the maximum value of a directional derivative at $(-2, 0)$, and find the unit vector in the direction in which the maximum value occurs.

If $\nabla f \neq 0$ at P , then among all possible directional derivatives of f at P , the derivative in the direction of ∇f at P has the largest value. This value is $|\nabla f|$ at P .

$$\nabla f(x, y) = [f_x(x, y), f_y(x, y)] \implies \nabla f(x, y) = [2xe^y, x^2 e^y]$$

The Gradient of f at $(-2, 0)$ is

$$\nabla f(-2, 0) = [-4, 4], \quad \text{and} \quad |\nabla f| = 4\sqrt{2}$$

This maximum occurs in the direction of $\nabla f(-2, 0)$. The unit vector in this direction is

$$\mathbf{u} = \frac{\nabla f}{|\nabla f|} = \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$$

A heat-seeking particle is located at the point $(2, 3)$ on a flat metal plate whose temperature at a point (x, y) is $T(x, y) = 10 - 8x^2 - 2y^2$. Find an equation for the trajectory of the particle if it moves continuously in the direction of maximum temperature increase.

A heat-seeking particle is located at the point $(2, 3)$ on a flat metal plate whose temperature at a point (x, y) is $T(x, y) = 10 - 8x^2 - 2y^2$. Find an equation for the trajectory of the particle if it moves continuously in the direction of maximum temperature increase.

Assume that the trajectory is represented parametrically by the equations

$$x = x(t), \quad y = y(t), \quad \text{at } t = 0 \text{ the particle is at } (2, 3)$$

Because the particle moves in the direction of maximum temperature increase, its direction of motion at time t is in the direction of the gradient of $T(x, y)$, and hence its velocity vector $\mathbf{v}(t)$ at time t points in the direction of the gradient. $\mathbf{v}(t) = k \nabla T$

$$\left[\frac{dx}{dt}, \frac{dy}{dt} \right] = k[-16x, -4y]$$

$$\left[\frac{dx}{dt}, \frac{dy}{dt} \right] = k[-16x, -4y]$$

$$\Rightarrow \frac{dx}{dt} = -16kx \text{ and } \frac{dy}{dt} = -4ky$$

$$\frac{dy}{dx} = \frac{y}{4x} \text{ with } y(2) = 3$$

$$\frac{dy}{y} = \frac{dx}{4x} \Rightarrow \ln y = \frac{1}{4} \ln x + \ln c$$

$$y = cx^{\frac{1}{4}}, y(2) = 3 \Rightarrow \boxed{y = 3 \left(\frac{x}{2} \right)^{\frac{1}{4}}}$$

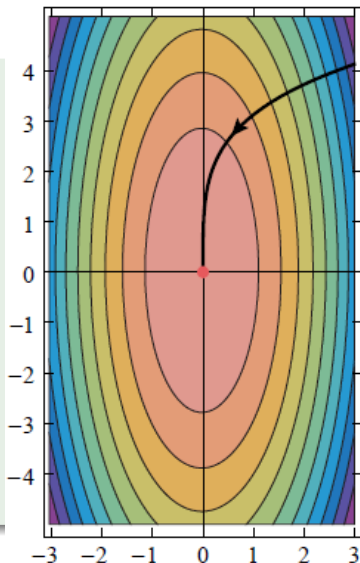
$$\left[\frac{dx}{dt}, \frac{dy}{dt} \right] = k[-16x, -4y]$$

$$\Rightarrow \frac{dx}{dt} = -16kx \text{ and } \frac{dy}{dt} = -4ky$$

$$\frac{dy}{dx} = \frac{y}{4x} \text{ with } y(2) = 3$$

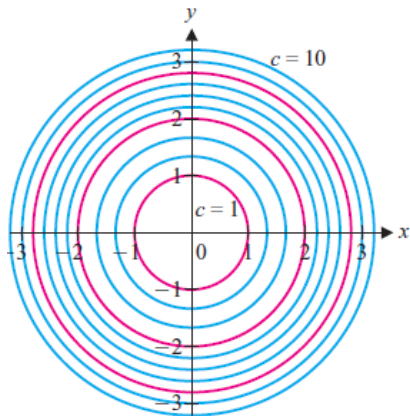
$$\frac{dy}{y} = \frac{dx}{4x} \Rightarrow \ln y = \frac{1}{4} \ln x + \ln c$$

$$y = cx^{\frac{1}{4}}, y(2) = 3 \Rightarrow y = 3 \left(\frac{x}{2} \right)^{\frac{1}{4}}$$

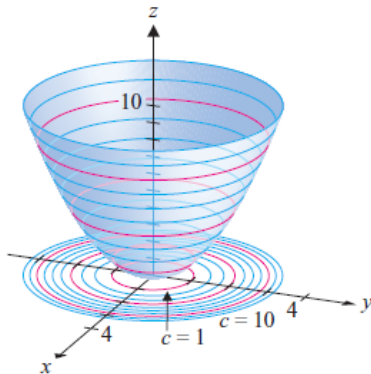


Level Curves or Contours

Level curves are traces in xy -planes $z = c$ of the surface $z = f(x, y)$.

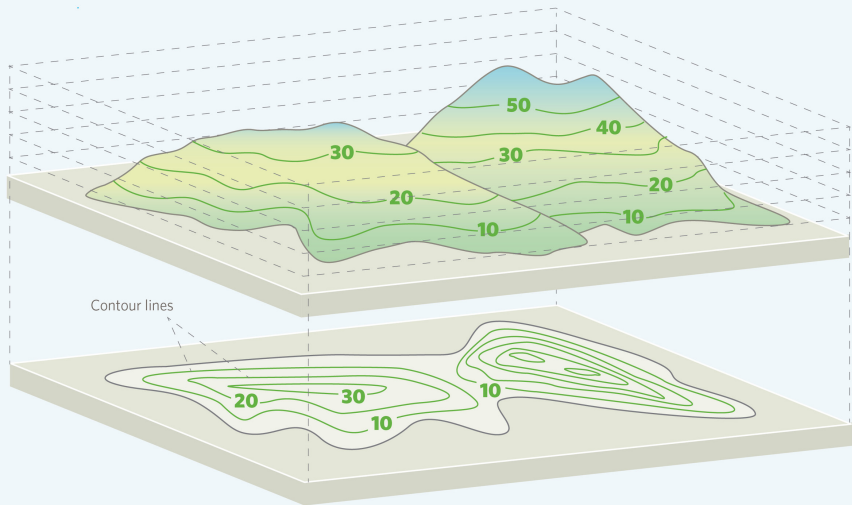


Contour plot of $g(x, y) = x^2 + y^2$



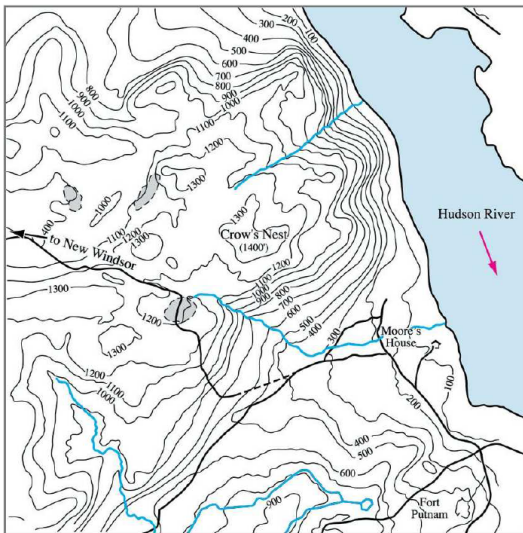
$z = x^2 + y^2$

Level Curves or Contours



The study of the land's surface shape, including hills, mountains, valleys, rivers, and craters. These features make up an area's terrain.

Contours and Gradient



Contours along the Hudson River in New York show streams, which follow paths of steepest descent, running perpendicular to the contours.

Partial Derivatives

The partial derivative of a function with respect to a variable is the derivative taken while keeping other variables constant.

For a function $f(x, y)$:

$\frac{\partial f}{\partial x}$ measures the rate of change of f with respect to x , with y constant.

Partial Derivatives - Example

Example: For the function $f(x, y) = x^2 + y^3$:

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 3y^2$$

- $\frac{\partial f}{\partial x}$: Derivative with respect to x , treating y as constant.
- $\frac{\partial f}{\partial y}$: Derivative with respect to y , treating x as constant.

Gradient of a Function

Definition: The gradient of a scalar function $f(x, y, z)$ is a vector that points in the direction of the steepest increase in f .

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

Each component is the partial derivative with respect to each variable.

Gradient - Example

For $f(x, y) = 3x^2 + 4y^2$:

$$\frac{\partial f}{\partial x} = 6x, \quad \frac{\partial f}{\partial y} = 8y$$

$$\nabla f = (6x, 8y)$$

- The gradient vector shows the direction of steepest increase in f at each point (x, y) .

Definition: The Hessian matrix of a function $f(x, y)$ is a square matrix of second-order partial derivatives:

$$H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}.$$

Purpose:

- Determines the local curvature of $f(x, y)$.
- Helps classify critical points as maxima, minima, or saddle points.

Determinant of the Hessian:

$$D = \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2.$$

Critical Points for $f(x, y)$

The point (a, b) is a critical point of the function $f(x, y)$ if (a, b) is in the domain of f and either

$$\frac{\partial f}{\partial x}(a, b) = \frac{\partial f}{\partial y}(a, b) = 0$$

or one or both of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ do not exist at (a, b) .

If $f(x, y)$ has a local extremum at (a, b) , then (a, b) must be a critical point of f .

Conditions for Extrema Using the Hessian

At a Critical Point (x_0, y_0) :

Local minimum

$$D > 0 \text{ and } \frac{\partial^2 f}{\partial x^2} > 0$$

Local maximum

$$D > 0 \text{ and } \frac{\partial^2 f}{\partial x^2} < 0$$

Saddle point $D < 0$

Inconclusive $D = 0$

Applications:

- Used in optimization problems.
- Crucial in machine learning, economics, and physics.

Locate and classify all critical points for $f(x, y) = 2x^2 - y^3 - 2xy$.

Step 1: Find Partial Derivatives

$$\frac{\partial f}{\partial x} = 4x - 2y, \quad \frac{\partial f}{\partial y} = -3y^2 - 2x$$

Step 2: Solve for Critical Points

$$\frac{\partial f}{\partial x} = 0 = \frac{\partial f}{\partial y}$$

$$4x - 2y = 0 \Rightarrow y = 2x,$$

$$-3y^2 - 2x = 0 \Rightarrow -12x^2 - 2x = 0.$$

$$x = 0, y = 0 \text{ and } x = -\frac{1}{6}, y = -\frac{1}{3}.$$

Critical points: $(0, 0)$ and $(-\frac{1}{6}, -\frac{1}{3})$.

Locate and classify all critical points for $f(x, y) = 2x^2 - y^3 - 2xy$.

Step 1: Find Partial Derivatives

$$\frac{\partial f}{\partial x} = 4x - 2y, \quad \frac{\partial f}{\partial y} = -3y^2 - 2x$$

Step 2: Solve for Critical Points

$$\frac{\partial f}{\partial x} = 0 = \frac{\partial f}{\partial y}$$

$$4x - 2y = 0 \Rightarrow y = 2x,$$

$$-3y^2 - 2x = 0 \Rightarrow -12x^2 - 2x = 0.$$

$$x = 0, y = 0 \text{ and } x = -\frac{1}{6}, y = -\frac{1}{3}.$$

Critical points: $(0, 0)$ and $(-\frac{1}{6}, -\frac{1}{3})$.

Step 3: Second Partial Derivatives

$$f_{xx} = 4, \quad f_{yy} = -6y, \quad f_{xy} = -2$$

Hessian determinant:

$$H = f_{xx}f_{yy} - (f_{xy})^2$$

At

$$(0, 0): H = (4)(-6(0)) - (-2)^2 = -4$$

$$H < 0 \Rightarrow \text{Saddle point}$$

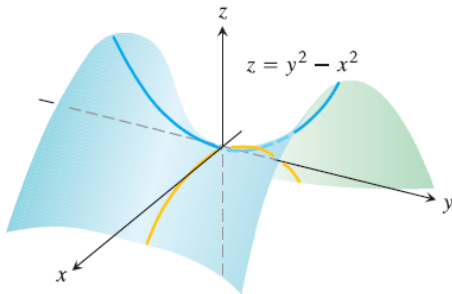
At $(-\frac{1}{6}, -\frac{1}{3}): H = 4$

$$f_{xx} > 0, \& H > 0 \Rightarrow \text{Local minimum.}$$

Final Classification of Critical Points

- $(0, 0)$: Saddle point.
- $(-\frac{1}{6}, -\frac{1}{3})$: Local minimum.

Find the local extreme values (if any) of $f(x, y) = y^2 - x^2$.



$$f_x = -2x, f_y = 2y \quad f_x = 0 = f_y \implies (x, y) = (0, 0)$$

$H = f_{xx}f_{yy} - (f_{xy})^2 < 0$ is a saddle point

$$f_{xx} = -2, f_{yy} = 2, f_{xy} = 0 \implies H = -4 < 0$$

Locate all relative extrema and saddle points of the function:

$$f(x, y) = 4xy - x^4 - y^4.$$

Step 1: Compute 1st-Order Partial Derivatives

$$f_x = 4y - 4x^3, \quad f_y = 4x - 4y^3.$$

$$x = (x^3)^3 = x^9 \implies x(x^8 - 1) = 0.$$

$$x = 0, x = \pm 1,$$

$$y = 0, y = \pm 1$$

Critical points:

- Set $f_x = 0$:
 $4y - 4x^3 = 0 \implies y = x^3.$
- Set $f_y = 0$:
 $4x - 4y^3 = 0 \implies x = y^3.$
- Solve $y = x^3$ and $x = y^3$:

Critical points:

$$(0, 0), \quad (1, 1), \quad (-1, -1).$$

Step 2: Compute Second-Order Partial Derivatives

$$f_{xx} = -12x^2, \quad f_{yy} = -12y^2, \quad f_{xy} = 4.$$

Hessian determinant:

$$H = f_{xx}f_{yy} - (f_{xy})^2 = (-12x^2)(-12y^2) - 4^2 = 144x^2y^2 - 16.$$

Step 3: Analyze Critical Points

Point	f_{xx}	f_{yy}	f_{xy}	H	Classification
(0, 0)	0	0	4	-16	Saddle point
(1, 1)	-12	-12	4	128	Relative maximum
(-1, -1)	-12	-12	4	128	Relative maximum

Relative maxima:

$$(1, 1) \quad \text{and} \quad (-1, -1).$$

Saddle point:

$$(0, 0).$$

Hessian Matrix

The Hessian matrix of a function is a square matrix of second-order partial derivatives. It provides information on the curvature of f .

For a function $f(x, y)$, the Hessian matrix H is:

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

Hessian Matrix - Example

For $f(x, y) = x^3 + 3xy + y^2$:

$$\frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial y^2} = 2, \quad \frac{\partial^2 f}{\partial x \partial y} = 3$$

$$H = \begin{bmatrix} 6x & 3 \\ 3 & 2 \end{bmatrix}$$

- The Hessian matrix helps determine the behavior of f near critical points.